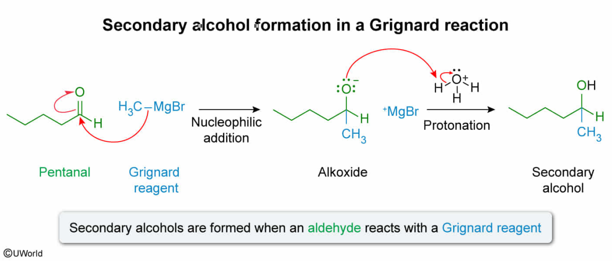
Which statement correctly describes an approach to synthesize a secondary alcohol?

- A. React pentanal with methylmagnesium bromide, followed by an acidic workup. [23%]
- B. React formaldehyde with methylmagnesium bromide, followed by an acidic workup. [17%]
- C. React pentanone with methylmagnesium bromide, followed by an acidic workup. [44%]
- D. React cyclohexanone with methylmagnesium bromide, followed by an acidic workup. [13%]

Correct
24% answered correctly

Explanation:



Ketones and aldehydes can be converted to alcohols using different reagents, including reducing agents and **organometallic reagents**. Grignard reagents are used in carbon-carbon bond-forming reactions between its alkyl group and an electrophilic molecule. When the electrophile is the carbonyl carbon of a ketone or aldehyde, the Grignard reagent causes a nucleophilic addition to the carbonyl, generating an alkoxide.

Protonation of the alkoxide during an acidic workup results in an alcohol, and the Grignard reagent adds one alkyl group to the electrophilic carbon. Therefore, the **type of alcohol formed** (ie, primary, secondary, or tertiary) contains **one more alkyl group than** the **number of alkyl groups** bonded to the **electrophilic carbonyl** used in the reaction:

- Reaction with formaldehyde (CH_2O) (zero alkyl groups) leads to formation of a primary alcohol
- Reaction with aldehydes (one alkyl group) leads to formation of a secondary alcohol
- Reaction with ketones (two alkyl groups) leads to formation of a tertiary alcohol

The question asks which statement describes an approach to form a secondary

alcohol. All the choices describe a reaction with methylmagnesium bromide, a Grignard reagent. Because secondary alcohols have **two** alkyl groups bonded to the hydroxyl group, an **aldehyde** must be the electrophile in the Grignard reaction. Aldehydes are named using the suffix –*al*. Therefore, the reaction between **pentanal** and methylmagnesium bromide, followed by an acidic workup, results in the formation of a secondary alcohol.

(Choice B) Formaldehyde is a unique aldehyde where the only carbon atom in the molecule is the carbonyl carbon. Therefore, zero alkyl groups are found in formaldehyde and reaction with a Grignard reagent, followed by acid, results in a *primary* alcohol.

(Choices C and D) The suffix –*one* indicates pentanone and cyclohexanone are ketones (each has *two* alkyl groups). Therefore, a *tertiary* alcohol forms when either of these reacts with a Grignard reagent, followed by an acidic workup.

Educational objective:
Ketones and aldehydes can be transformed into alcohols by nucleophilic addition of an alkyl group to the carbonyl by an organometallic reagent (eg, Grignard reagent), followed by acidic workup. The type of alcohol formed (ie, primary, secondary, or tertiary) contains one more alkyl group than the number of alkyl groups bonded to the carbonyl reactant.

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